Muneebullah Nawaz

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Education

King Abdullah University of Sciences and Technology (KAUST)| |Thuwal, Saudi Arabia *Sep 2025 –Present*
Doctor of Philoshpy in Mechanical Engineering, **GPA 3.75, KAUST Fellowship**
Advisor: Professor Peter Schmid

The Ohio State University (OSU)| |Columbus, OH, United States *August 2023 – July 2025*
Master of Science in Mechanical Engineering, **GPA 3.94, Graduate Research Assistant**
Advisor: Professor Shawn Midlam-Mohler
Projects:
- Simulation-Driven Design Optimization and Automation for Cordwood-Fueled Room Heaters (SPRING' 24- Present)
- Capstone Project – CFD of battery pack for thermal loads analysis
- Control-oriented Thermal Load Characterization Model for the Electric School Bus

King Abdullah University of Sciences and Technology (KAUST)| |Thuwal, Saudi Arabia *Jan 2021 – Aug 2023*
Master of Science in Mechanical Engineering, **GPA 3.75, KAUST Fellowship**
Thesis: Vertical Wind Tunnel Design via Integrated Numerical and Statistical Modeling (*Data-Driven Modeling and Computational Fluid Dynamics)*
Advisor: Professor Tadd Truscott

National University of Sciences and Technology (NUST)| Islamabad, Pakistan *September 2013 – September 2017*
Bachelor of Engineering in Aerospace, **GPA 3.85**, **President's Gold Medal for 1st position in the batch** Thesis:
Conceptual Design and Mockup development of a Hexagonal Micro-Satellite *(System Design)*
Advisor: Professor Ali Sarosh

Skills

Software: MATLAB, Simulink, Python, C++, OpenVSP**,** Catia, SolidWorks, Ansys (Mechanical, Fluent), LabVIEW
Hardware: Wind Tunnel Testing and Aerodynamic Experimentation**,** Structural Impact Testing
Professional Competencies: Leadership, Project Management, and Team Coordination, Technical Writing: Design Proposals, Reports, and Presentations
Languages: Urdu (Native), English (Advanced)

Professional Research Experience (4 years)

Turkish Aerospace Industries (TAI) *Feb 2022 – Jun 2022*
Aero-elasticity and Dynamics Load Engineer *(Aeroelasticity and Dynamic Load Analysis)*
- Contributed to the development of a hybrid model for Continuous Turbulence Load Analysis, gaining hands-on experience with aircraft aero-elastic behavior.
- Assisted in the analysis of aero-elastic behavior of aircraft components under various load conditions.
- Engaged in initial stages of compliance with FAA certification requirements for proposed aircraft design.

Aviation Research Indigenization and Development *October 2017 – May 2021*
Team Lead – Aircraft Design Optimization and Integration Group *(Simulation System Modeling and Design Optimization)* • Developed a modeling and simulation environment for aircraft conceptual design, reducing design evaluation time by 75%.
- Led the development of a novel stochastic optimization algorithm, successfully applied in real-world scenarios.
- Automated the design process for aircraft modeling, improving efficiency and accuracy in design evaluations.

Publications

1. Analysis of the Vortex-Induced Vibrations on Chimney and Effect of Variable Stiffness Guy Cable VortexInduced Vibrations on the Chimney Wake Region under Different Forcing Functions to Reduce the Vibrations of Chimney
2. Determination of Directional Stability using Torsional Pendulum Concept for Preliminary Design of Aerospace Vehicles
3. A control-oriented thermal load characterization model for the e-school bus: Auxiliary Load Analysis (SAE Technical Paper)
4. Advancing Data-Driven Turbulence Modeling with Transformer Neural Networks: A Comprehensive Review
5. Optimizing Vertical Wind Tunnel Design through Integrated Numerical and Statistical Modeling